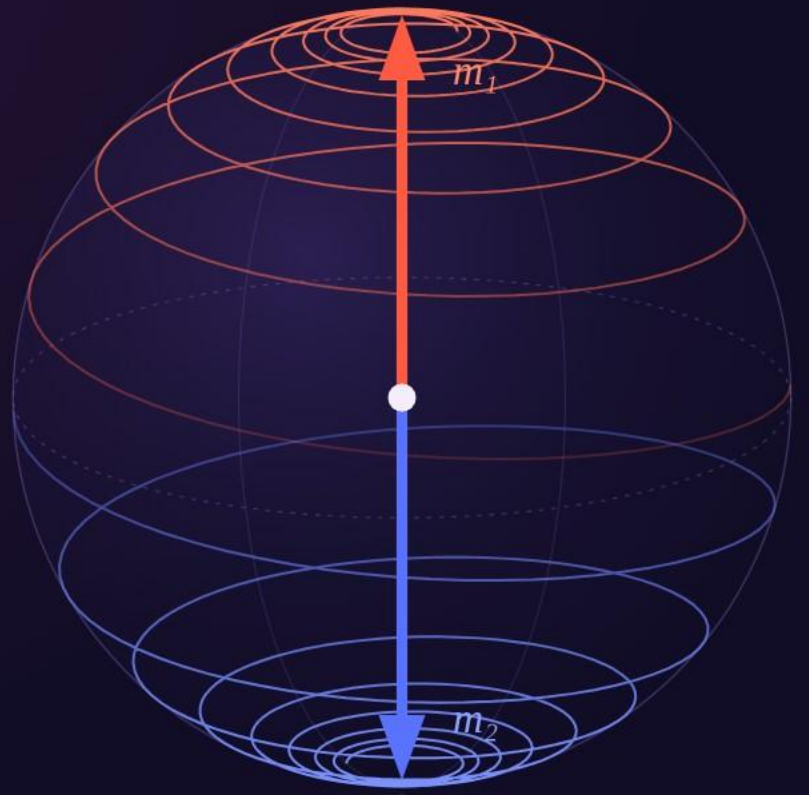
spinmate

AI AGENT FOR SPINTRONICS SIMULATION · VERSION 2.0

# Spintronics simulation, by conversation.

Describe a setup in plain language or provide a parameter file. SpinMate checks it with the solver's own parser, runs two-sublattice macrospin simulations and explains the result from the actual magnetization dynamics.




Michael Lianeris · github.com/Michaellianeris/Spinmate-AI-Agent · MIT license

# Outline

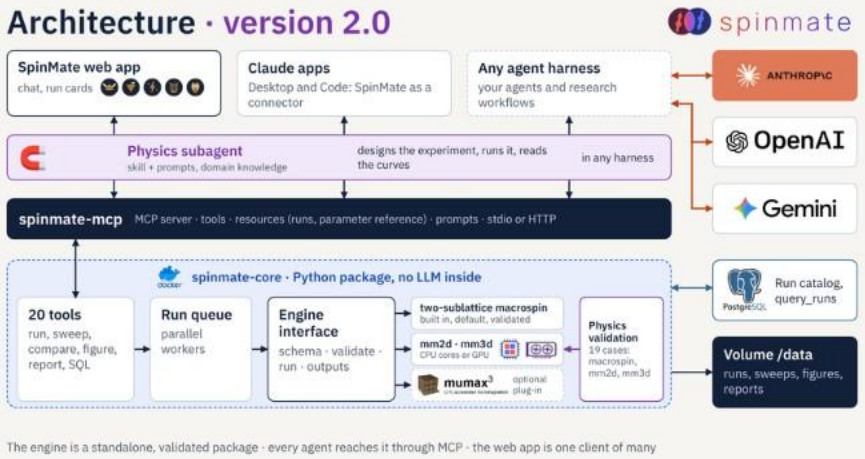

| | | |
|---|---|---|
| 01 | Version 2.0 | What changed: any agent through MCP, a core package, validation, engines |
| 02 | Architecture | How the clients, the MCP server, the core and the engines fit together |
| 03 | Models and value | Five personas over Claude, OpenAI and Gemini; runs, sweeps, figures, reports on real runs |
| 04 | Trusted physics | The validation suite and the engines: macrospin, mm2d and mm3d for skyrmions, spin waves and hopfions, mumax3 |
| 05 | Numbers and setup | Speed, tests, tools, the codebase, and how to run it |

# What’s new in Version 2.0

## Version 1.0 (today)




## Version 2.0 (proposal)




### Any agent

An MCP server: Claude Code, Claude Desktop and the web app use the same tools

### spinmate-core

The engine as a package with no LLM inside, and its own command line

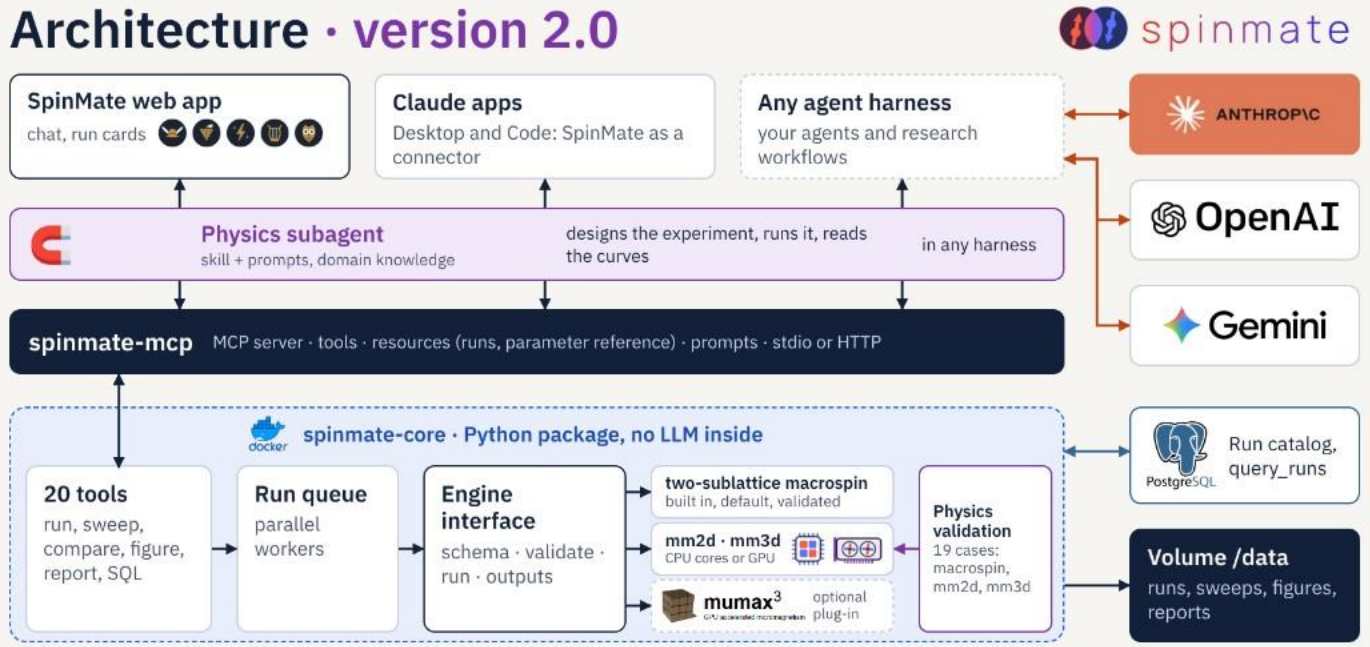
### Validated physics

9 analytic cases on every change; 4 solver issues found and fixed

### Engines

macrospin and mm2d (2D skyrmions) built in, mumax3 as a plug-in

# Architecture




## Clients

The web app, Claude Code, Claude Desktop, any MCP agent

## spinmate-mcp

19 tools, resources and prompts, with a physics subagent

## spinmate-core

Run queue, engines (macrospin, mm2d, mm3d, mumax3), validation, catalog, /data

# Use it from any agent

### Claude Code

```
claude mcp add --transport http \
  spinmate http://localhost:8421/mcp
```

### Without Docker

```
python -m spinmate_mcp          # stdio
python -m spinmate_mcp --http   # :8422
```

Claude Desktop: the same server over stdio, in claude_desktop_config.json. The repository brings a physics subagent and skill for Claude Code.

**19 tools**
Validate, run, sweep, wait, get the plot, compare, figure, report, SQL on the catalog

**Resources**
spinmate:// parameters, physics guide and scales, validation, runs, examples

**Prompts**
design_experiment, find_threshold, interpret_run

**Same safety**
Large requests return needs_confirmation; the agent asks you first

# Five personas, one router



## Hermes

**Quick and economical**

Claude Haiku, GPT nano / mini, Gemini Flash-Lite



## Dionysus

**Balanced, the default**

Claude Sonnet, GPT-5.x, Gemini Flash



## Zeus

**Deep, most capable**

Claude Opus, OpenAI o-series, Gemini Pro



## Apollo

**Specialised and new**

Claude Fable / Mythos, new models



## Athena

**The guide**

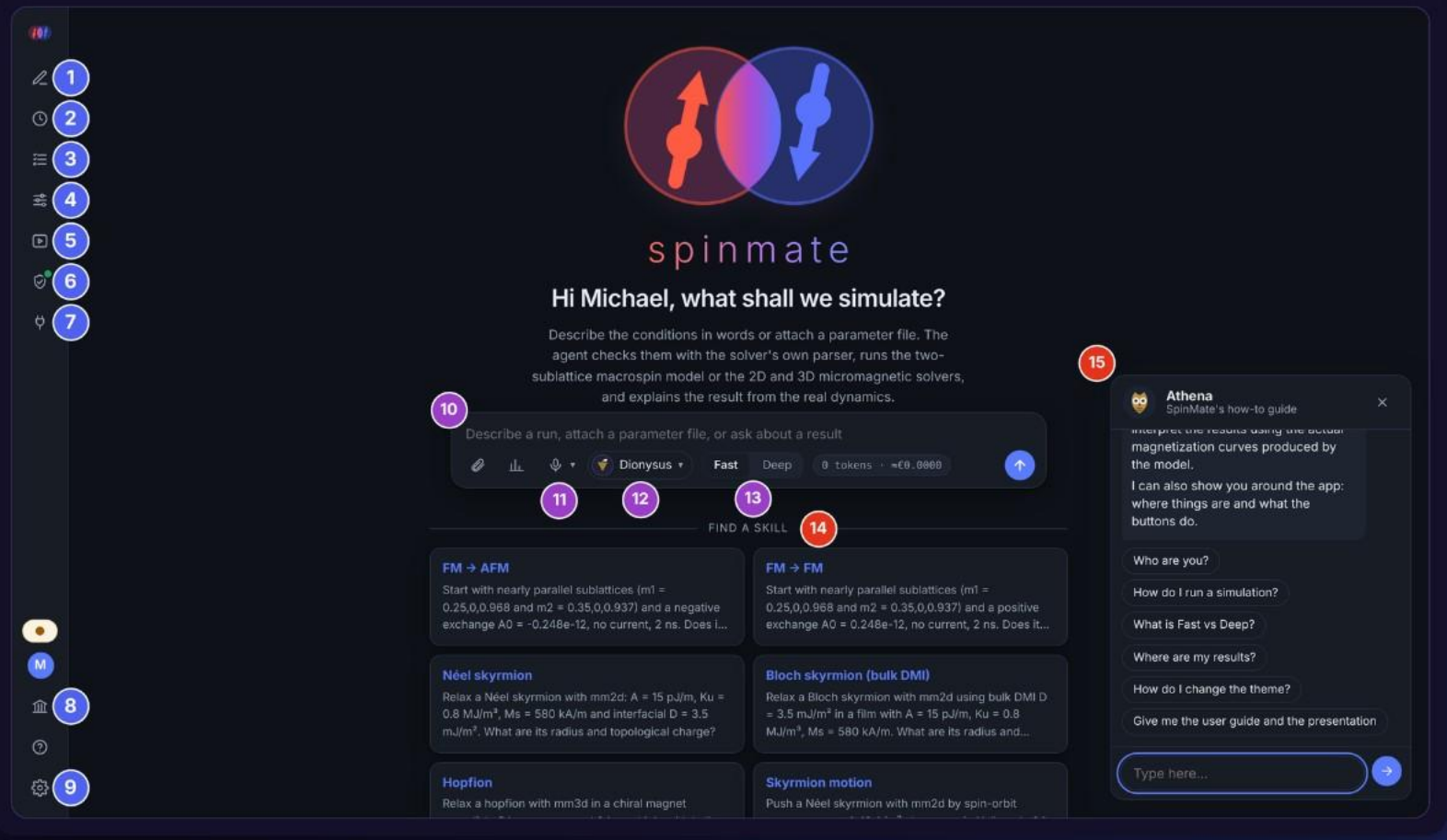
How to use the app; built-in answers offline

ON AUTO, EVERY REQUEST IS SCORED OVER THE PERSONA'S MODELS

$$\text{score} = \text{capability} + \text{bonus} - \lambda(\text{persona}, \text{request}) \times \log_{10}(\text{price} / \text{cheapest}) - \text{shortfall}$$

The strongest model wins unless a cheaper one is clearly enough. Every answer shows the model, the reason and the cost in €, at the daily ECB rate.

# A tour of the home page




## LEFT BAR

1. New conversation (Ctrl+K)
2. Conversation history
3. Runs panel and catalog (Ctrl+/)
4. Default parameters
5. Run setup, no AI needed
6. Physics validation
7. Connect to Claude (MCP)
8. Model gods, tools guide
9. Settings: language, theme, API keys

## MESSAGE BOX

10. Ask, attach a .txt, or the sweep form
11. Voice input
12. Persona: the model level
13. Fast or Deep; tokens and € per provider

## START HERE

14. Find a skill: ready questions
15. Athena: how-to, guide and slides

# What SpinMate gives you

BY HAND

1. Edit the parameter file
2. Run the solver and wait
3. Write a plotting script
4. Read the curves for switches and settling
5. Collect parameters and figures into a write-up

WITH SPINMATE

1. One sentence, or the same .txt file
2. Every value checked by the solver’s parser
3. Run card with the sphere, curves and metrics
4. Answer written from the computed metrics
5. IEEE figure and PDF report on request

**Same solver, same physics: the agent removes the steps around it, not the model.**

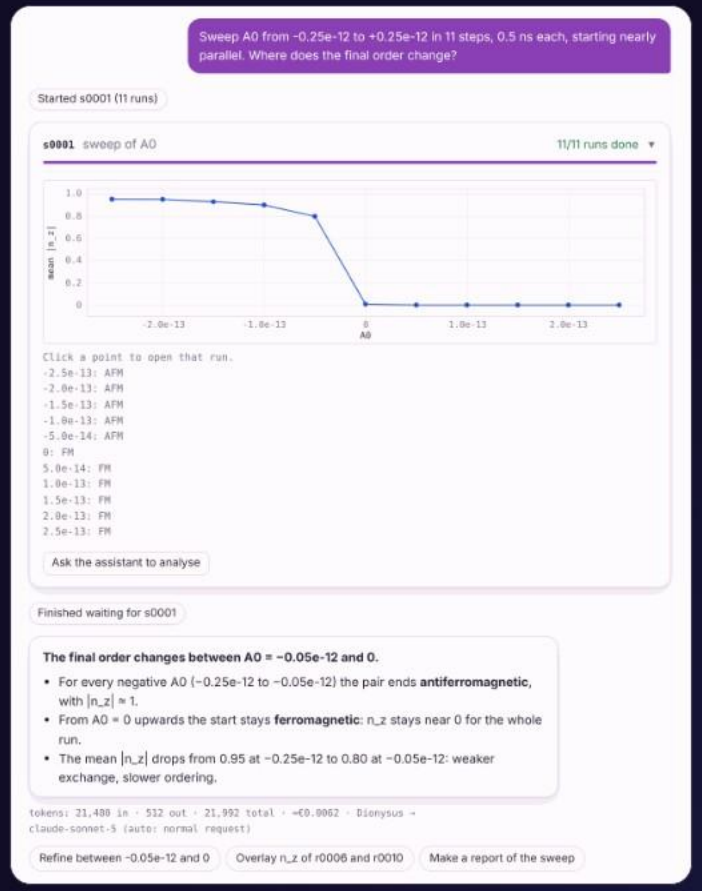
# One question, one run




## Behind one question

1. The router picks a model for the chosen persona and request.
2. The model calls validate_parameters, then start_simulation.
3. The run queue starts the solver; the card shows live progress.
4. wait_for_run returns the metrics, and the answer is written from them.

# Sweeps, thresholds and continued runs



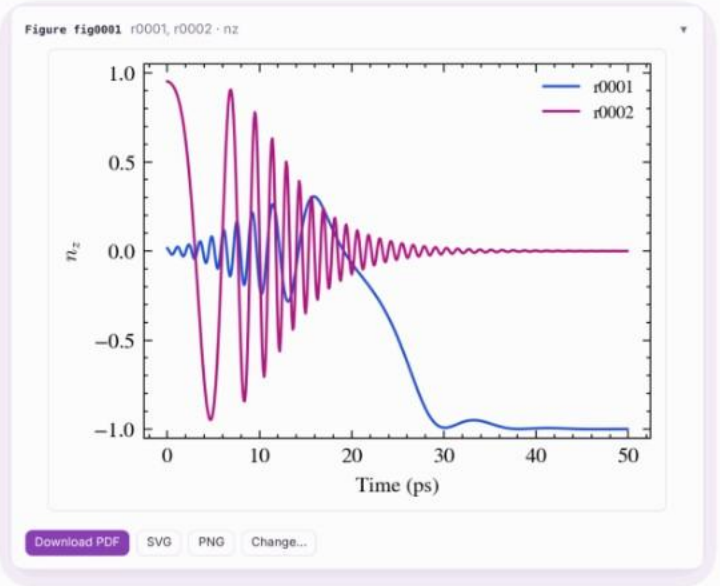

*“Sweep A0 from −0.25e-12 to +0.25e-12 in 11 steps, 0.5 ns each. Where does the final order change?”*

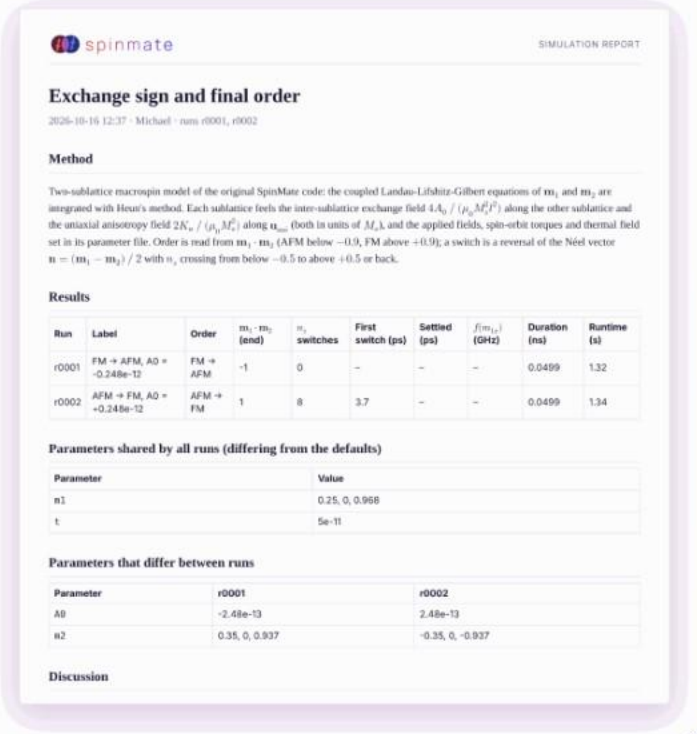
RESULT

## The order changes between A0 = −0.05e-12 and 0: AFM below, FM from 0 upwards.

- Runs go in parallel; click a point to open that run.
- Threshold search: bracketing sweeps narrow the interval.
- Continue run keeps the clock, so AC fields and the FeRh cycle do not restart.

# Figures and reports, ready to share





spinmate

SIMULATION REPORT

**Exchange sign and final order**

2026-10-16 12:37 · Michael · runs r0001, r0002

**Method**

Two-sublattice macrospin model of the original SpinMate code: the coupled Landau-Lifshitz-Gilbert equations of $\mathbf{m}_1$ and $\mathbf{m}_2$ are integrated with Heun's method. Each sublattice feels the inter-sublattice exchange field $4A_0 / (\mu_0 M_s^2 l^2)$ along the other sublattice and the uniaxial anisotropy field $2K_u / (\mu_0 M_s^2)$ along $\mathbf{u}_{ani}$ (both in units of $M_s$), and the applied fields, spin-orbit torques and thermal field set in its parameter file. Order is read from $\mathbf{m}_1 \cdot \mathbf{m}_2$ (AFM below $-0.9$, FM above $+0.9$); a switch is a reversal of the Néel vector $\mathbf{n} = (\mathbf{m}_1 - \mathbf{m}_2) / 2$ with $n_z$ crossing from below $-0.5$ to above $+0.5$ or back.

**Results**

| Run | Label | Order | $\mathbf{m}_1 \cdot \mathbf{m}_2$ (end) | $n_z$ switches | First switch (ps) | Settled (ps) | $f(m_{1z})$ (GHz) | Duration (ns) | Runtime (s) |
|---|---|---|---|---|---|---|---|---|---|
| r0001 | FM → AFM, A0 = -0.248e-12 | FM → AFM | -1 | 0 | – | – | – | 0.0499 | 1.32 |
| r0002 | AFM → FM, A0 = +0.248e-12 | AFM → FM | 1 | 8 | 3.7 | – | – | 0.0499 | 1.34 |

**Parameters shared by all runs (differing from the defaults)**

| Parameter | Value |
|---|---|
| n1 | 0.25, 0, 0.968 |
| t | 5e-11 |

**Parameters that differ between runs**

| Parameter | r0001 | r0002 |
|---|---|---|
| A0 | -2.48e-13 | 2.48e-13 |
| n2 | 0.35, 0, 0.937 | -0.35, 0, -0.937 |

**Discussion**

## Figures

IEEE, Nature and science styles, as PDF, SVG and PNG.

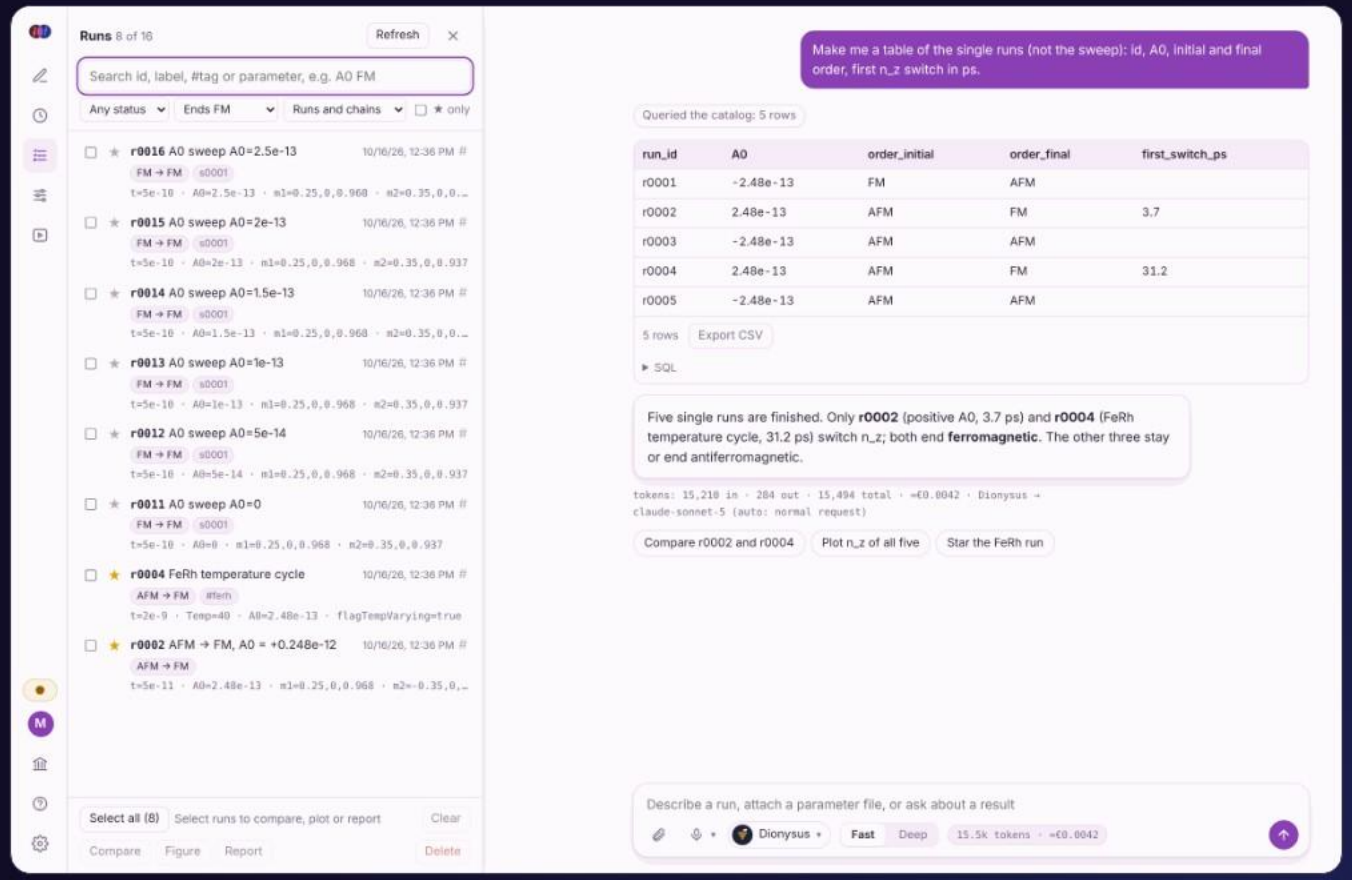
## Reports

Method, results, parameters and figures, with LaTeX math and the SpinMate header.

## Export

Print to PDF, or download the HTML or a Markdown and figures zip.

# Every run in one catalog




## One row per run

Postgres in Docker, SQLite in the Windows app: parameters, summary and the links between runs.

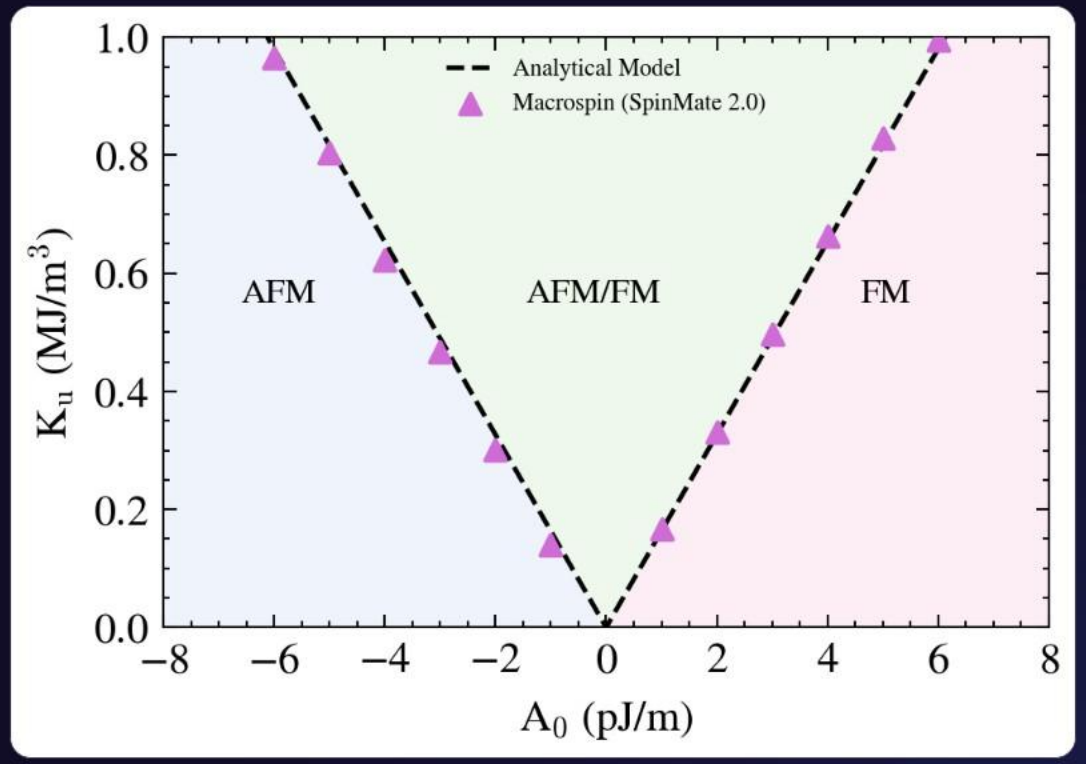
## Questions across runs

The model writes one read-only SELECT; the result comes back as a table with Export CSV.

## Runs panel

Search by id, label, #tag or parameter; filter, star, compare, plot or report.

# Validated physics




Measured AFM/FM boundary (12 points) on $|Ku| = 2|A0|/l^2$, within 2–6 %

| Case | Expected | Measured |
|---|---|---|
| $\|m_1\| = \|m_2\| = 1$ | 0 | 2e-16 |
| Same input, same trajectory | 0 | 0 |
| Energy at α = 0 | conserved | drift 1.6e-6 |
| Kittel mode | 5.62 GHz | 5.61 GHz |
| AFM resonance | 67.3 GHz | 67.3 GHz |
| Time-step convergence | h = 1e-13 s | 0.03 % |
| Gilbert damping α·ω | 0.3525 /ns | 0.3524 /ns |
| AFM/FM boundary | $2\|A0\|/l^2$ | at ±10 % |
| Thermal field σ | Temp_Ampl | within 3 % |

**4 solver issues fixed:** anisotropy counted twice, Ku(t) amplitude, RK4 weights, thermal noise. Runs at T = 0 stay bit-identical.

python -m spinmate_core.validation · 19 cases: macrospin, mm2d, mm3d · runs on every pull request

# Four engines, one interface

BUILT IN · DEFAULT

## macrospin

- Two coupled LLG equations, $m_1$ and $m_2$
- Exchange, anisotropy, SOT, temperature
- Sweeps and continuation
- Runs on any CPU in seconds

BUILT IN · 2D, 3D

## mm2d · mm3d

- SpinMate's own 2D and 3D micromagnetics
- Skyrmions (FM, AFM, SAF) and their motion, spin waves, hopfions
- Relaxation or LLG dynamics
- No GPU, seconds per film

PLUG-IN · GPU

## mumax3

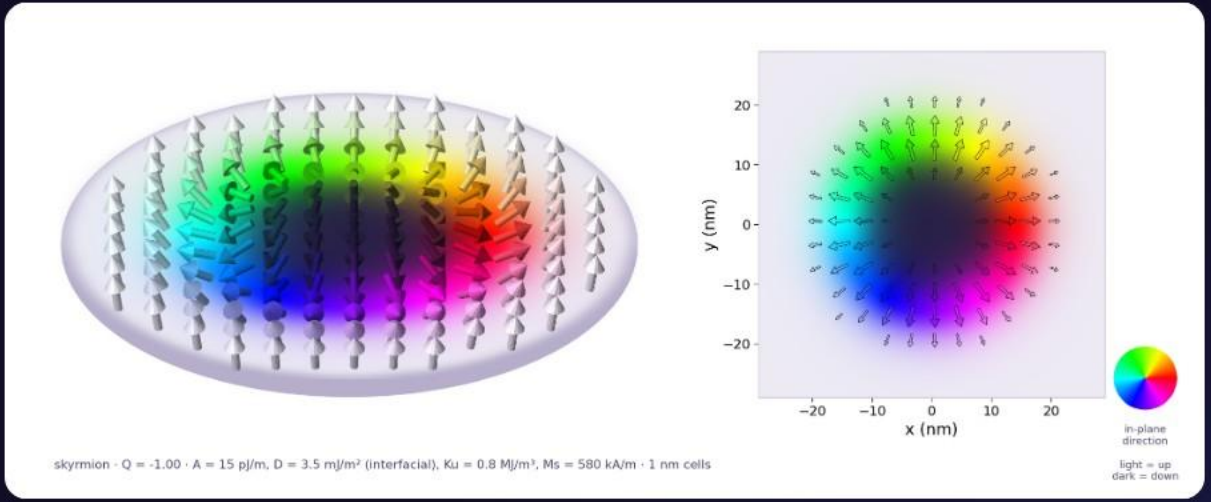
- GPU micromagnetics on a grid
- SpinMate lines or an .mx3 script
- Run card shows ⟨m⟩(t)
- Needs NVIDIA GPU, MUMAX3_BIN

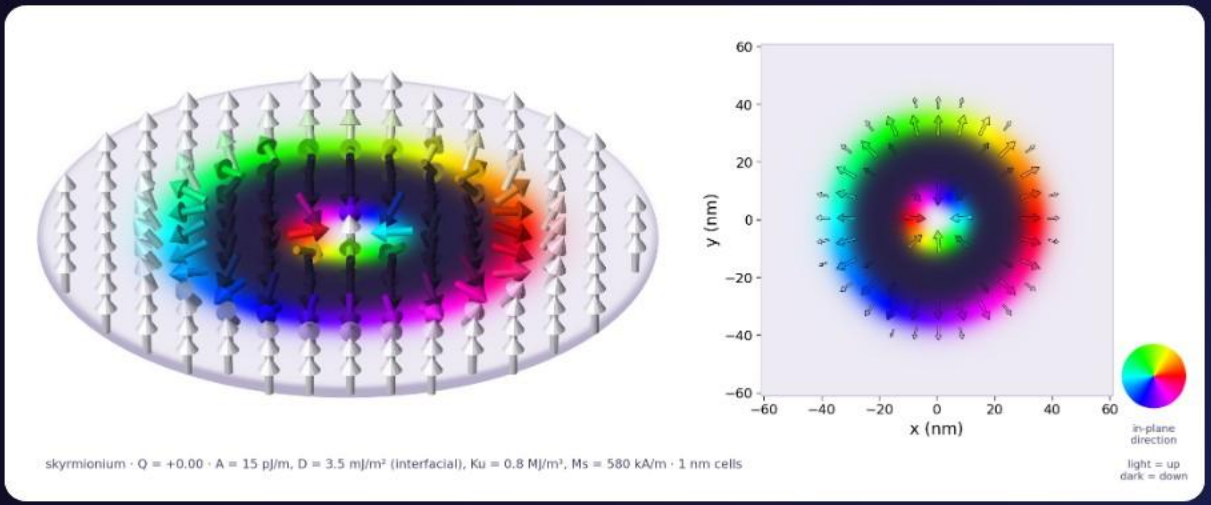
```
[project.entry-points."spinmate.engines"]
mumax3 = "spinmate_mumax3:Mumax3Engine"
```

One interface: parameter_schema, validate, run_command, outputs. mumax3 (GPL v3) runs as a separate program, so SpinMate stays MIT.

Cite mumax3: Vansteenkiste et al., AIP Advances 4, 107133 (2014)

# Skyrmions with mm2d · built-in 2D micromagnetics




A thin film one cell thick: exchange, anisotropy, interfacial or bulk DMI, field and thin-film demag. Relax to the stable texture, or run LLG with a field pulse.

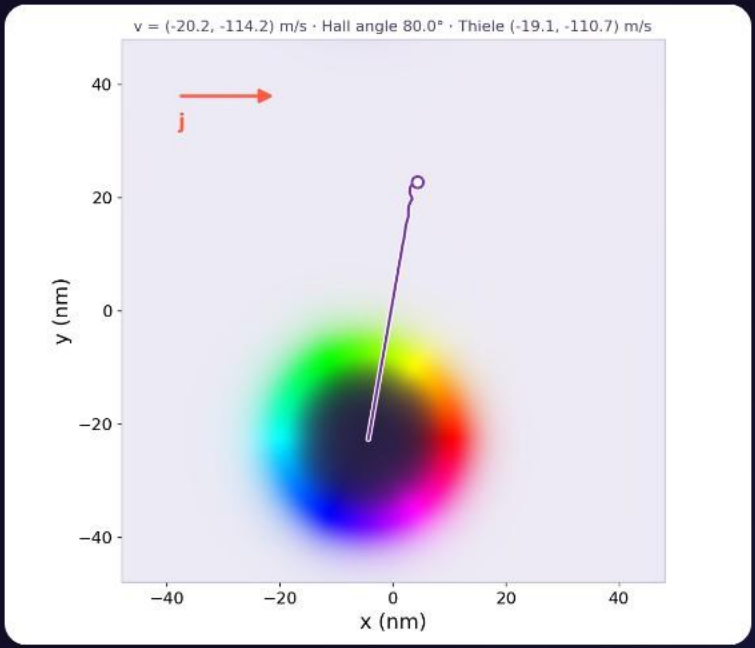
| Check | Result |
|---|---|
| Wall energy $4\sqrt{(A \cdot K_{eff})} - \pi\lvert D\rvert$ | within 0.3 % |
| Wall width $\sqrt{(A/K_{eff})}$ | within 1 % |
| Kittel frequency | within 0.2 % |
| Energy at $\alpha = 0$ | drift < 1e-4 |
| Skyrmion / skyrmionium Q | −1.00 / 0.00 |

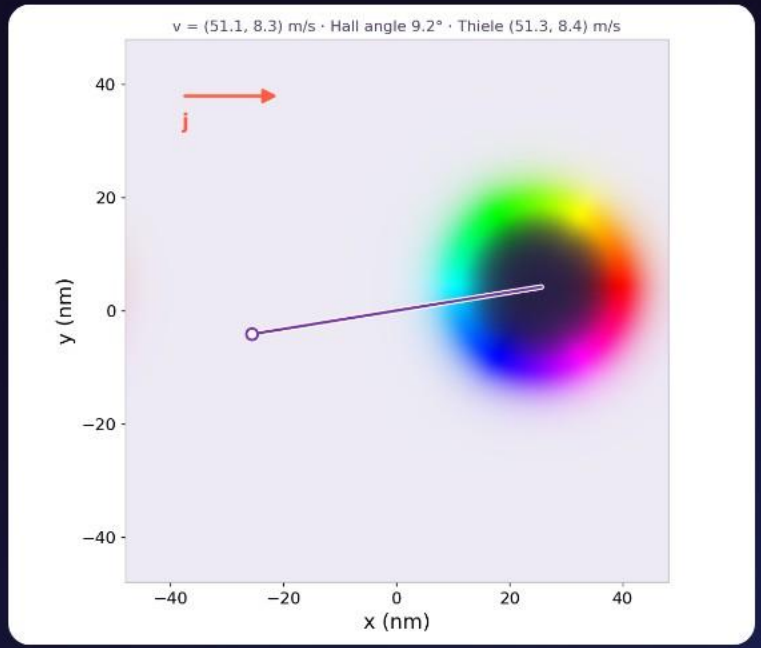
**5 s**
128 × 128 skyrmion, CPU

**14.4 nm**
skyrmion radius, D = 3.5 mJ/m²

Colour: in-plane direction; light = up, dark = down.

# Skyrmions on the move · SOT and STT




## 116 m/s

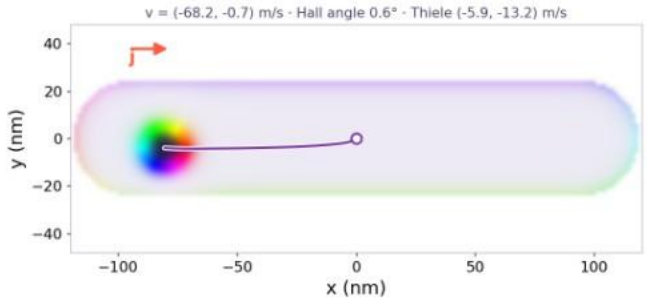
SOT, 1e12 A/m², deflected 80°

## v = u

STT with β = α, within 1 %

Every run is compared with the Thiele equation for the relaxed skyrmion: within 1 % for STT and 3 % for SOT.

# Any shape from a PNG · a skyrmion racetrack



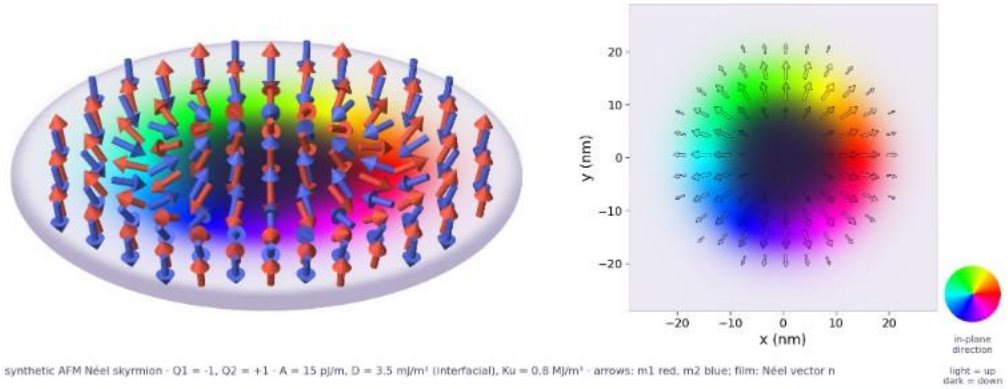

**68 m/s**
along the track; 14.5 m/s in an open film

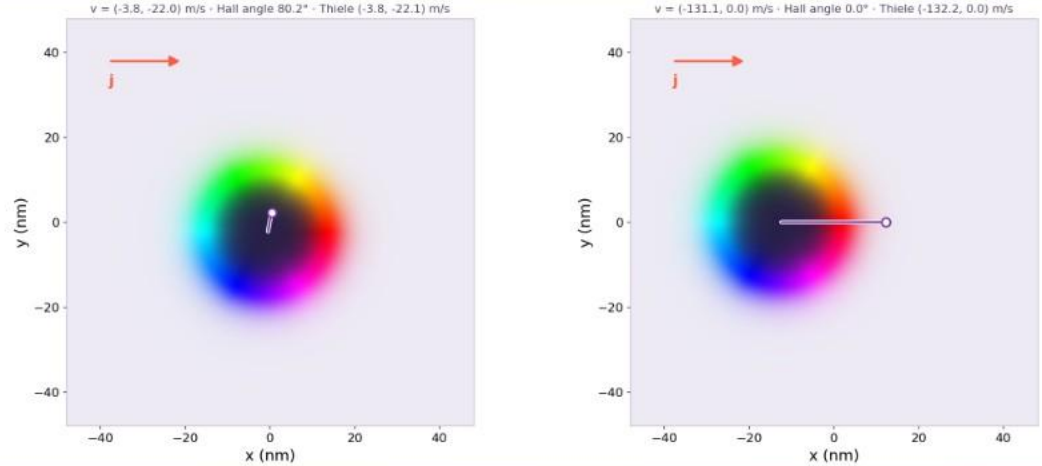
**1 %**
edge tilt vs sin θ = Dδ/2A

Drop a PNG on the chat: dark pixels become the film. Edges carry the DMI boundary condition, so they tilt and keep the skyrmion on the track.

engine="mm2d" · mask=racetrack | disk | strip | your.png

# Antiferromagnetic skyrmions · no Hall effect



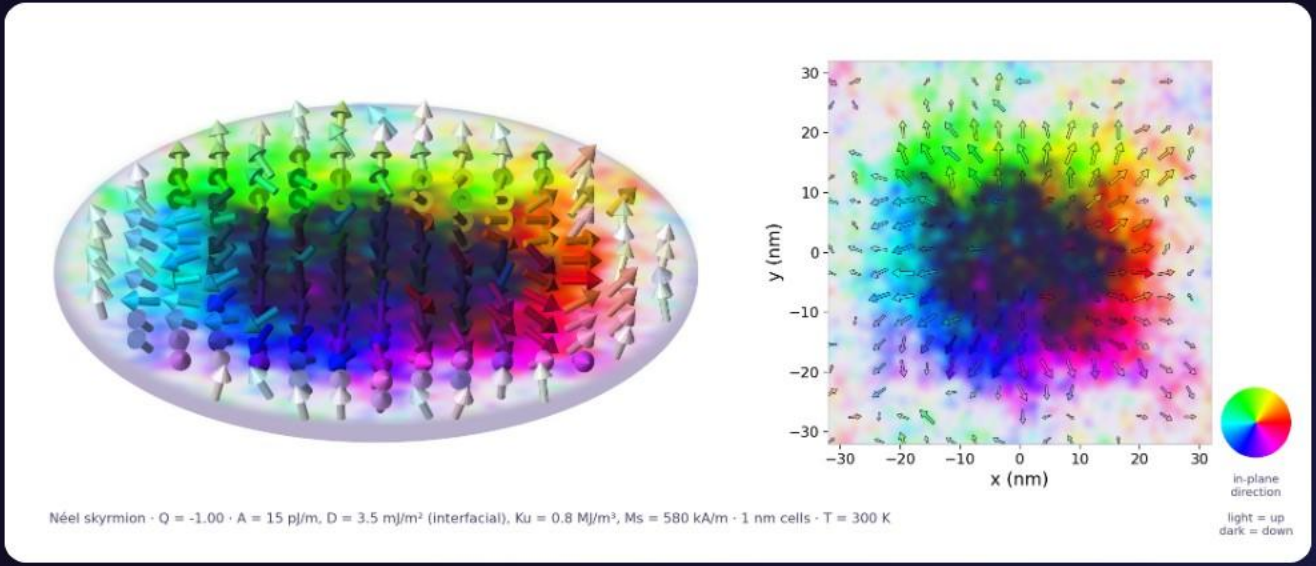



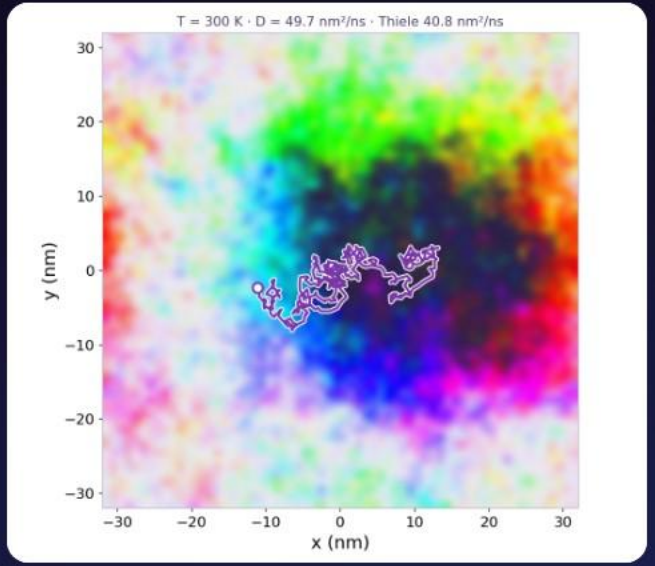

## 131 m/s, 0°

synthetic AFM, against 22 m/s and 80° in one layer

Two layers coupled antiparallel (RKKY) carry skyrmions of opposite charge: the Magnus forces cancel. Thiele within 1 %.

# Skyrmions at 300 K · stochastic LLG




**Boltzmann**

$\langle m_z \rangle$ of free spins exact for α = 0.1 and 0.5

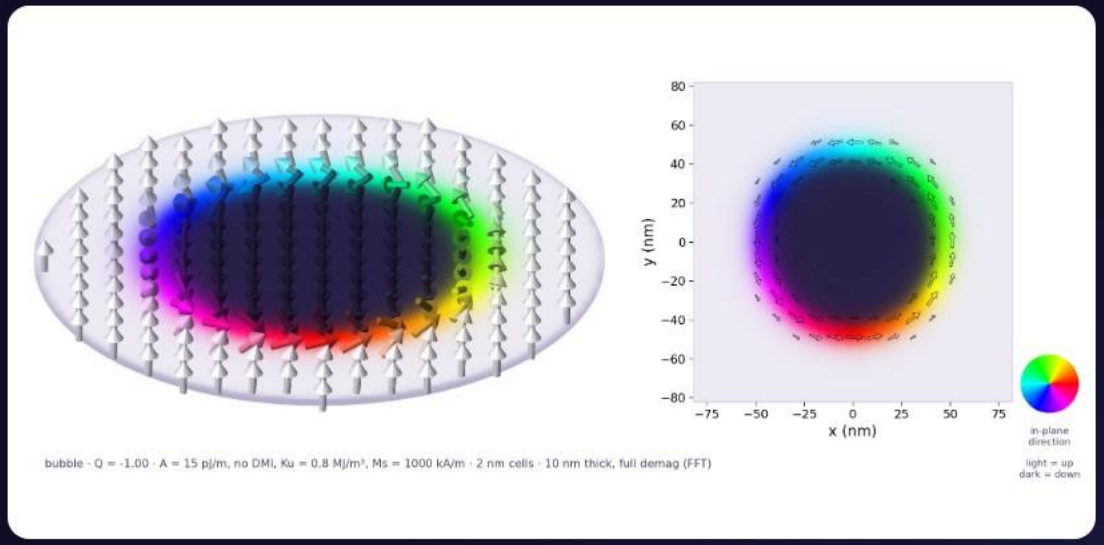
**50 nm²/ns**

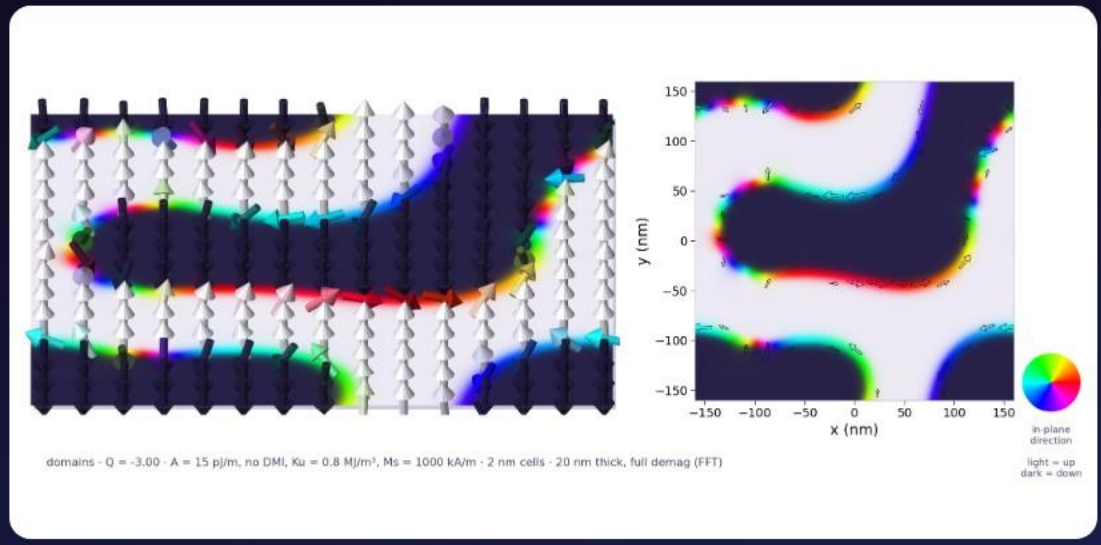
skyrmion diffusion, Thiele 41 (one trajectory)

Brown's thermal field with fixed Heun steps; works with currents, pulses and both lattices.

engine="mm2d" · mode=dynamics · Temp · thermal_dt

# Bubbles and domains · long-range demag by FFT



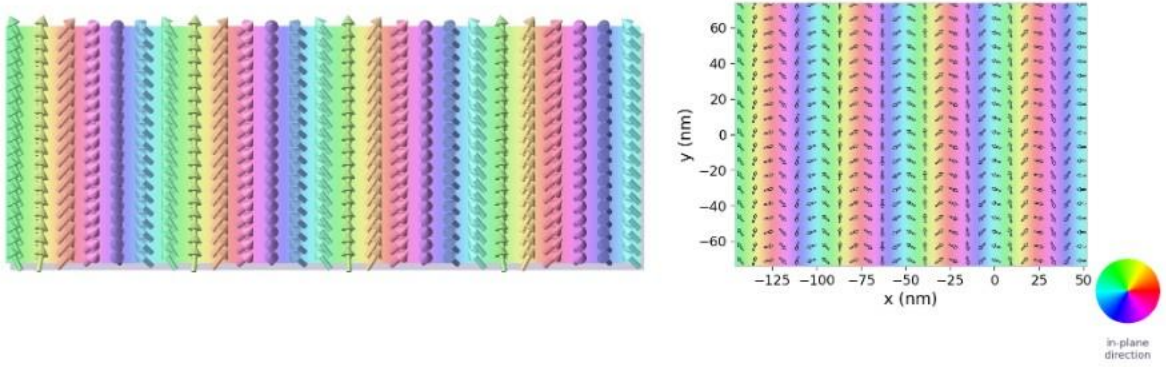




**0.4 %**
Néel-wall stray-field energy vs $\mu_0 M_s^2 d \cdot \ln 2/\pi$

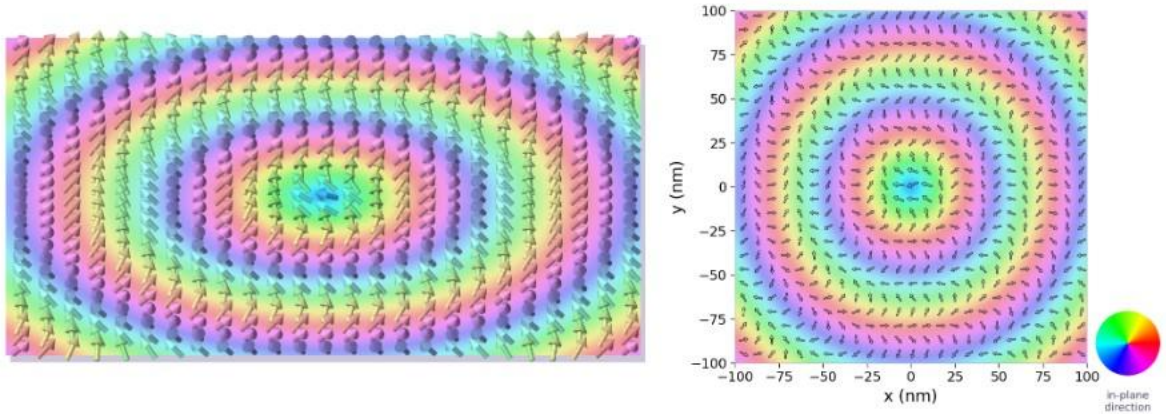
**$10^{-6}$**
platelet demag factor vs Aharoni

Newell tensor, FFT convolution, free or periodic films. Thicker films, dipolar bubbles without DMI and maze domains that the local field cannot give.

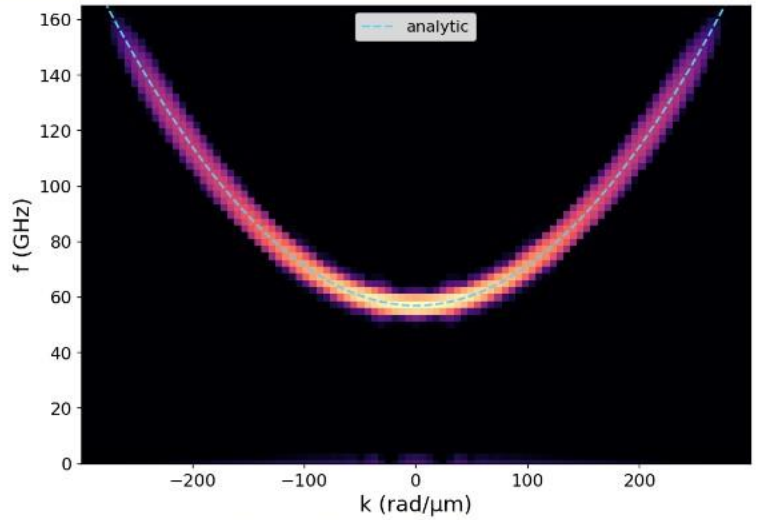
# Spin waves with mm2d · from an antenna




spin waves from a stripe antenna · sine 80 GHz · t = 300 ps · λ = 49.3 nm (theory 49.6 nm) · gap 56.8 GHz · precession enlarged to show the phase




spin waves from a dot antenna · sine 80 GHz · t = 250 ps · λ = 48.4 nm (theory 49.6 nm) · gap 56.8 GHz · precession enlarged to show the phase




mx(x, t) along y = 0 · 512 × 2 nm cells · line: $\gamma_0/2\pi \cdot (H + H_{K,eff} + 2A\,k^2/\mu_0 M_s)$ · median mismatch 0.7 %

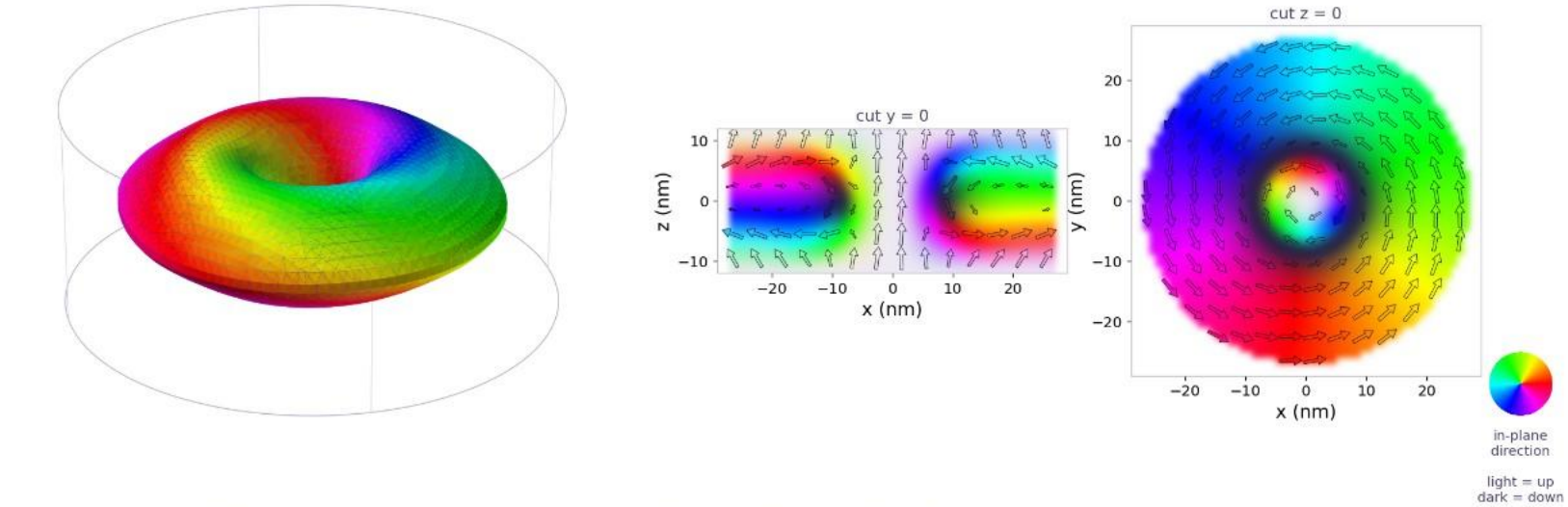
**49.3 nm**
λ at 80 GHz, theory 49.6 nm

**< 1 %**
f(k) against theory

Stripe or dot antenna, sine or broadband sinc pulse, absorbing edges.
Colour: phase of the precession, enlarged.

engine="mm2d" · mode=dynamics · excite=stripe | dot · pulse=sine | sinc · absorb

# Hopfions with mm3d · built-in 3D micromagnetics




hopfion · Hopf index H = -0.89 · surface m_z = 0, coloured by in-plane direction · disk 54 nm × 24 nm, p = 24 nm · B = 0.3 T

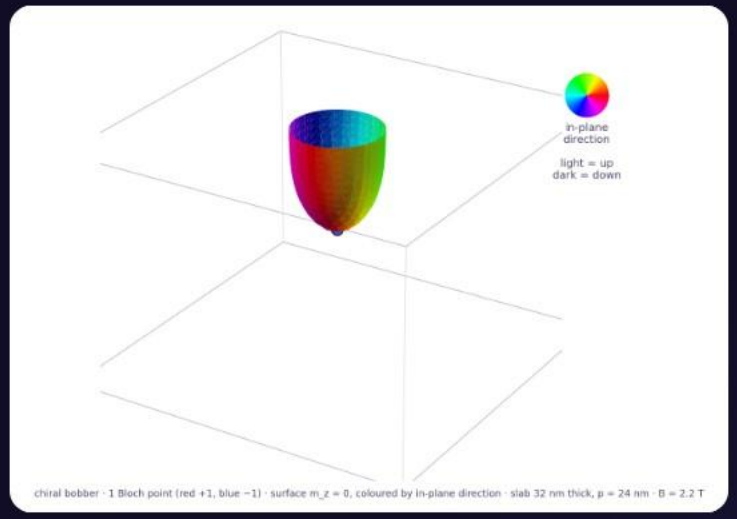
**|H| = 1**
Hopf index, −0.89 on the grid

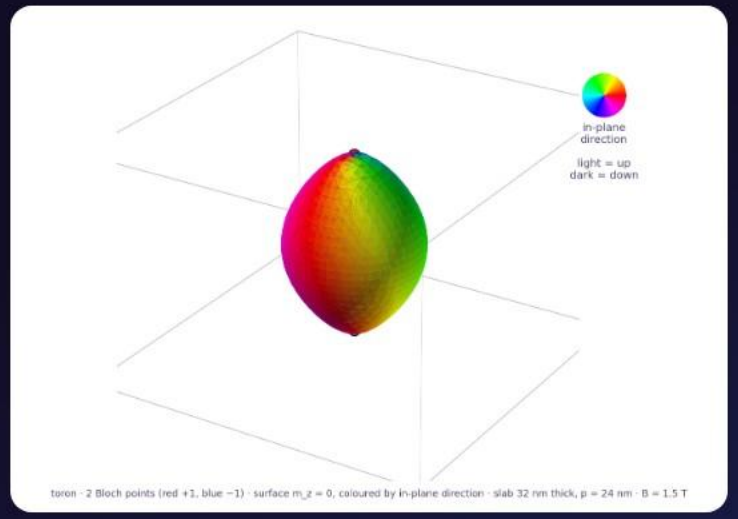
**73 s**
58 × 58 × 24 cells, CPU

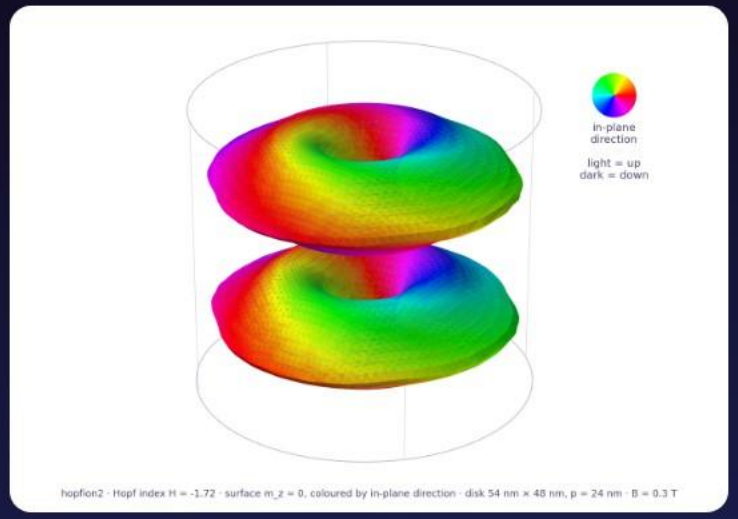
Chiral magnet nanodisk ~2p across and p thick, surfaces pinned up. The $m_z = 0$ surface is a torus; its colour turns once around the ring: the linking that makes a hopfion.

engine="mm3d" · texture=hopfion | skyrmion_tube · geometry=disk | slab · surfaces=pinned

# Bobbers, torons, H = 2 · Bloch points found



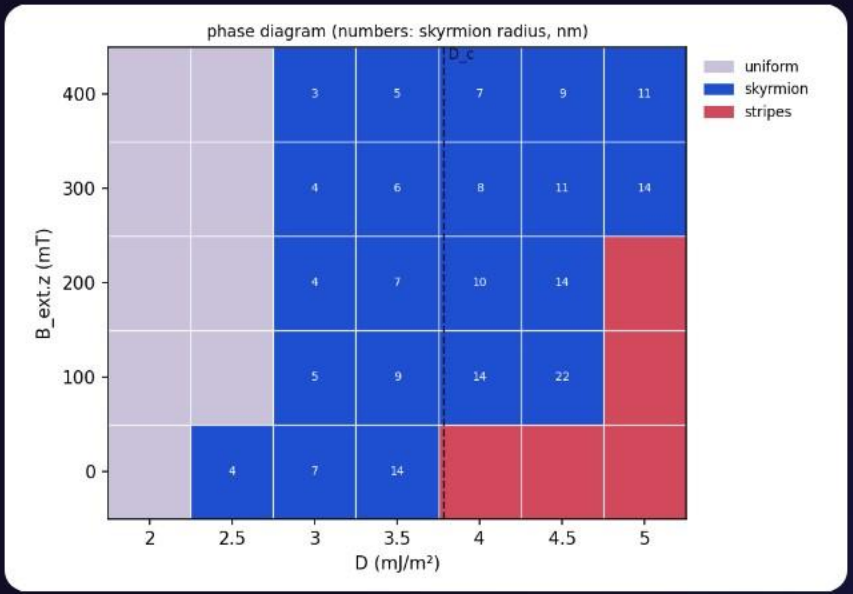

**±1**
Bloch-point charge from the solid angle of m

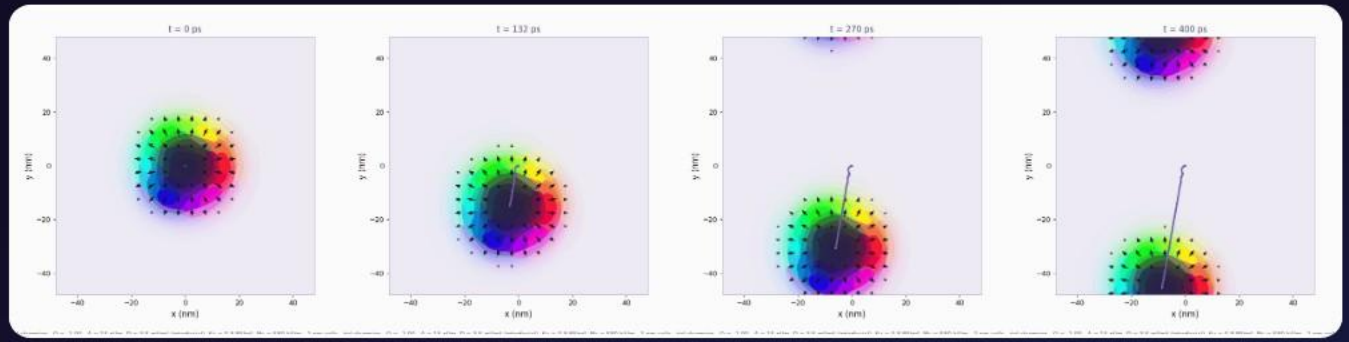
**2 × 0.86**
Hopf index of two stacked hopfions

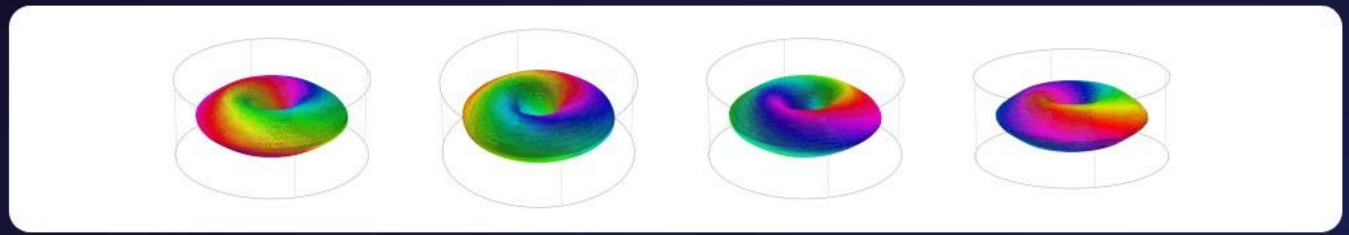
A bobber ends half a period below a free surface; a toron is closed by +1 and −1 between pinned surfaces. Dots mark the Bloch points.

engine="mm3d" · texture=chiral_bobber | toron | hopfion2

# Phase diagrams · and animations



**$D_c = 3.78$**

mJ/m²: stripes start where $4\sqrt{(A \cdot K_{eff})}/\pi$ says

The macrospin sweep machinery now runs mm2d and mm3d: one parameter gives the radius, two give a phase diagram. Spin waves, moving skyrmions and 3D textures come back as GIF and MP4.

start_sweep · engine=mm2d · parameter2 · animation.gif · rotate.gif

# SpinMate in numbers

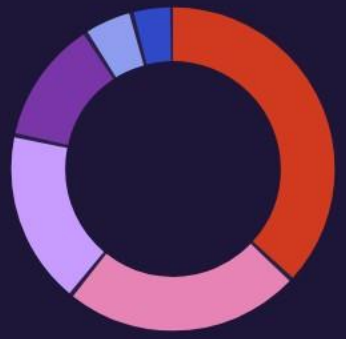
**1.3 s**
to simulate 50 ps (500 steps), run r0001

**51 s**
for an 11-run A0 sweep, 0.5 ns each

**182**
automated tests, all passing

**16**
eval cases through the real agent and solver

### Codebase by part · 13,314 lines



- **37 %** web UI
- **24 %** core, MCP and plug-ins
- **18 %** agent and web server
- **13 %** tests and evals
- **5 %** solver, validated
- **4 %** desktop app and guide

### The 19 tools by job



- **5** inputs, validation, engines
- **5** runs
- **3** sweeps
- **2** continue and combine
- **4** compare, figure, report, SQL

Measured on the runs in this deck; lines of .py, .html and .js in the repository

# Runs on your machine

### Local by default

Listens on 127.0.0.1 only. Docker on Linux, macOS and WSL, or a Windows app.

### Your keys

In .env or the browser tab, never committed. Calls bill your own account; TOKEN_BUDGET sets a cap.

### Your data

Runs, figures, reports and accounts stay in /data. Only the optional welcome email leaves; MAIL_RELAY=off turns it off.

### Guard rails

Read-only SQL with a timeout; runs start only through the solver’s parser; big requests ask first, over MCP too.

# Get started

## Docker

```
cp .env.example .env
docker compose up --build
```

Add an Anthropic, OpenAI or Gemini key to .env, then open localhost:8421.

## Windows app

Download SpinMate-windows.zip from Releases, unzip and run SpinMate.exe. No Docker or Python needed; the catalog is SQLite.

## First start

Athena asks your name and, optionally, your email for the PDF guide. No key yet? Run setup runs a preset directly.

**From Claude Code** `claude mcp add --transport http spinmate http://localhost:8421/mcp`

spinmate **github.com/Michaellianeris/Spinmate-AI-Agent**

Free and open source under the MIT license · the user guide is web/spinmate-guide.pdf